

Fitness Tracker App – Project Report

## **CODTECH IT SOLUTIONS PRIVATE LIMITED**

**MERN Stack Web Development Internship — Project Report**

**FITNESS TRACKER APP PROJECT REPORT**

**TOPIC: MERN Stack Full-Stack Application – Fitness & Workout Tracking System**

|                          |                                                                           |
|--------------------------|---------------------------------------------------------------------------|
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| <b>Organization</b>      | Codtech IT Solutions Private Limited                                      |
| <b>Internship Period</b> | 03 June 2026 – 15 July 2026                                               |
| <b>Duration</b>          | 6 Weeks                                                                   |
| <b>Domain</b>            | MERN Stack Web Development                                                |
| <b>Project Type</b>      | Fitness Tracker App                                                       |
| <b>Tools / Language</b>  | React.js, Express.js, Node.js, MongoDB, Mongoose, Axios, React Router DOM |

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## 1. Overview

The Fitness Tracker App is a full-stack web application built using the MERN stack (MongoDB, Express.js, React.js, and Node.js). It allows users to manage their daily workout routines by creating, viewing, updating, and deleting exercise records through an intuitive and responsive interface. The backend exposes RESTful APIs for handling user and exercise data, while MongoDB provides reliable and scalable data storage.

The project demonstrates practical full-stack development concepts, including React's component-based architecture, Express REST APIs, MongoDB integration through Mongoose, and client-server communication using Axios. The frontend is built with React.js and communicates with the Express.js backend, while exercise and user information is persisted in MongoDB via Mongoose models.

## 2. Features

### 2.1 Workout Management

- Add new workout exercises
- Edit existing exercise records
- Delete workouts
- View all stored exercises
- Track workout duration
- Store exercise date

### 2.2 User Management

- Create new users
- Retrieve registered users
- Associate exercises with individual users

### 2.3 Backend Features

- RESTful API architecture
- Express.js routing
- MongoDB database integration
- Mongoose schema validation
- JSON request and response handling
- Modular backend structure

### 2.4 Frontend Features

- React component architecture
- Responsive user interface
- React Router navigation
- Axios for API communication
- Organized pages and reusable components

## 3. Tech Stack

### 3.1 Frontend

- React.js
- React Router DOM
- Axios
- JavaScript (ES6)
- HTML5
- CSS3

### 3.2 Backend

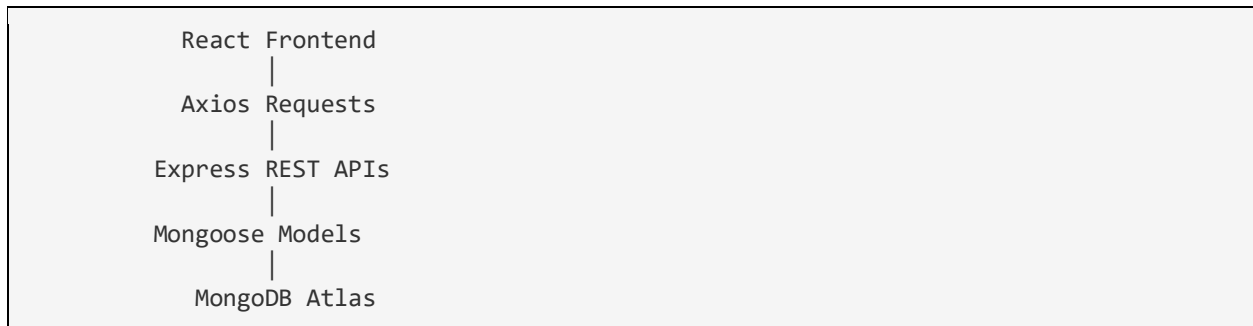
- Node.js
- Express.js
- MongoDB
- Mongoose

### 3.3 Development Tools

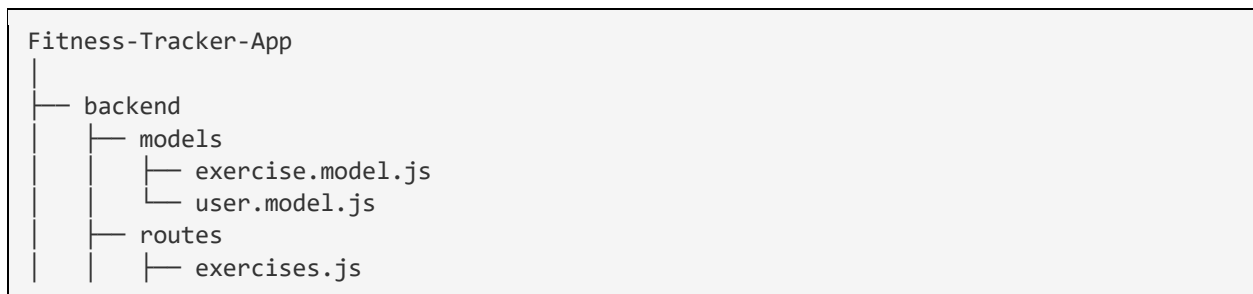
- Git & GitHub
- VS Code
- npm
- Nodemon
- Postman

## 4. System Architecture

The application follows a layered client-server architecture, as illustrated below:



## 5. Project Structure



```
├── users.js
├── package.json
├── package-lock.json
├── Procfile
├── server.js
├── client
│   ├── public
│   │   ├── images
│   │   ├── index.html
│   │   └── _redirects
│   ├── src
│   │   ├── components
│   │   ├── images
│   │   ├── pages
│   │   ├── App.js
│   │   └── index.js
│   ├── package.json
│   └── package-lock.json
├── README.md
├── LICENSE
└── .gitignore
```

## 6. Installation

### 6.1 Clone the Repository

```
git clone https://github.com/Yug912/Fitness-Tracker-App.git
cd Fitness-Tracker-App
```

### 6.2 Install Backend Dependencies

```
cd backend
npm install
```

### 6.3 Install Frontend Dependencies

```
cd ../client
npm install
```

## 7. Environment Variables

Create a .env file inside the backend folder with the following variables:

```
ATLAS_URI=mongodb+srv://username:password@cluster.mongodb.net/fitness_tracker
PORT=5000
```

## 8. Running the Application

### 8.1 Start the Backend Server

```
cd backend
npm start
# Runs at http://localhost:5000
```

### 8.2 Start the Frontend

```
cd client
npm start
# Runs at http://localhost:3000
```

## 9. REST API Endpoints

### 9.1 Users

#### Get All Users

```
GET /users
```

#### Create User

```
POST /users/add

{ "username": "Yug" }
```

### 9.2 Exercises

#### Get All Exercises

```
GET /exercises
```

#### Get Exercise by ID

```
GET /exercises/:id
```

#### Create Exercise

```
POST /exercises/add

{
  "username": "Yug",
  "description": "Chest Workout",
  "duration": 60,
  "date": "2026-07-09"
}
```

#### Update Exercise

```
POST /exercises/update/:id
```

### Delete Exercise

```
DELETE /exercises/:id
```

## 10. Database Schema

### 10.1 User Model

| Field    | Type   | Description                                |
|----------|--------|--------------------------------------------|
| username | String | Unique display name of the registered user |

### 10.2 Exercise Model

| Field       | Type   | Description                                 |
|-------------|--------|---------------------------------------------|
| username    | String | References the user who logged the exercise |
| description | String | Short description of the workout performed  |
| duration    | Number | Duration of the workout, in minutes         |
| date        | Date   | Date on which the exercise was performed    |

## 11. Application Workflow

```
graph TD
    A[User Opens Application] --> B[React Frontend]
    B --> C[Axios Sends Request]
    C --> D[Express Receives Request]
    D --> E[Route Handler]
    E --> F[MongoDB Database]
    F --> G[JSON Response]
```

↓  
React Updates UI

## 12. Learning Outcomes

This project provided hands-on experience with the following concepts:

- MERN Stack development end-to-end
- REST API design and implementation
- CRUD operations across frontend and backend
- MongoDB database integration with Mongoose
- React Router based navigation
- Component-based front-end architecture
- Client-server communication using Axios
- Express middleware usage
- Mongoose ODM for schema validation
- Responsive UI development

## 13. Screenshots

The following application screens were captured during testing and demonstration:

- Home Page
- Dashboard
- Exercise List
- Create Exercise
- Update Exercise
- Delete Exercise

*(Insert corresponding screenshots here before final submission.)*

## 14. Future Enhancements

- JWT-based authentication
- Login and registration system
- Password encryption
- User profiles
- Exercise categories
- Progress dashboard with weekly/monthly analytics
- Workout charts and calories-burned estimation
- Goal tracking and BMI calculator
- Dark mode and mobile optimization
- Docker deployment and CI/CD pipeline
- Unit and integration testing

## 15. Conclusion

The Fitness Tracker App successfully demonstrates the design and implementation of a complete MERN stack application. It covers backend REST API development, MongoDB data modelling, and a responsive React frontend, giving a well-rounded understanding of full-stack web development practices used in real-world projects.

## **16. Author**

**Yug Thakral**

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LinkedIn: <https://www.linkedin.com/in/yug-thakral-a38252317/>